

Experiment 16

Understand EDA, Quantitative and Qualitative analysis.

Program:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline

tips=sns.load_dataset('tips')
tips.head()

# Quantitative analysis
sns.displot(tips.total_bill,kde=True)
sns.displot(tips.total_bill,kde=False)

sns.jointplot(x=tips.tip,y=tips.total_bill)
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")

sns.pairplot(tips)

tips.time.value_counts()
sns.pairplot(tips,hue='time')
sns.pairplot(tips,hue='day')

sns.heatmap(tips.corr(numeric_only=True),annot=True)
sns.boxplot(tips.total_bill)
sns.boxplot(tips.tip)

# Qualitative analysis
sns.countplot(tips.day)
sns.countplot(tips.sex)

tips.sex.value_counts().plot(kind='pie')
tips.sex.value_counts().plot(kind='bar')
sns.countplot(tips[tips.time=='Dinner']['day'])
```

Output

Output:

Sample rows:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Time value counts:

Dinner	176
Lunch	68

Output plots:

Distribution plots, joint plots, pair plots, correlation heatmap, box plots, count plots, pie chart and bar chart are generated.